AI Vehicle Damage Classification | Bul Ins, Bulgaria

Expanded domain research • October 2026

1.1 Research question and scope

To what extent can a machine-learning image-classification model accurately classify visible vehicle damage from photographs to support initial motor-insurance claims triage at Bul Ins, while accounting for operational, legal, privacy, ethical and environmental constraints?

The proposed start-up product classifies visible damage (e.g., dent, scratch, crack, glass shatter, broken lamp and flat tire). It does not decide claim validity, compensation or total loss. A Bul Ins claims handler or inspector remains responsible for decisions. The six-class baseline is a project-specific experiment, not a verified description of Bul Ins production data.

1.2 Insurance and business environment

Bulgarian claims context

Bul Ins is a Bulgarian motor insurer. Its Autocasco general terms describe vehicle inspection and photographic documentation, reporting obligations, damage inspection, and compensation based on parts, labour and materials. According to Bul Ins's 2026 Autocasco general terms, policyholders must notify the insurer of most insured events within three working days of becoming aware of them. The insurer must pay compensation within 15 days after receiving all required documents, subject to the applicable claim conditions (Bul Ins, 2026).

The Bulgarian Financial Supervision Commission reported disputes about refusals and compensation amounts among common insurance complaints in Q3 2025. This supports the importance of clear, reviewable decisions, but does not prove that automated photo classification will reduce complaints (FSC, 2025).

Stakeholder evidence and unresolved problem validation

In the student interview, Bul Ins employees described manual data entry, saving and distributing documents, and repetitive checks. They did not

specifically identify photo-based damage classification as their main bottleneck. Therefore the operational value of the proposed classifier remains a hypothesis requiring a follow-up interview and process observation (Bul Ins stakeholder interview, project evidence).

Key stakeholders are claims handlers, vehicle inspectors, policyholders, Bul Ins management, IT/security staff, compliance and data-protection staff, and the start-up developer. Their needs may conflict: speed and cost savings must be balanced against fair treatment, accuracy and accountability.

1.3 Technology and data environment

Image-classification feasibility

Computer-vision methods have been explored for insurance triage, including earlier research on applying image analysis to vehicle claims (Natsev et al., 2007). The student baseline uses k-NN on image crops; this is a simple comparison point rather than a deployment-ready solution.

The exploratory analysis of CarDD identified uneven class representation. A higher overall accuracy may hide poor recognition of rare damage types. Evaluation should therefore include class-wise precision, recall, confusion matrices, confidence calibration and performance on genuinely unseen images. Data leakage between photos of the same damage or vehicle must be prevented (Project EDA and Iteration Zero evidence).

Data suitability and generalisation

Public datasets may differ from Bul Ins photographs in camera quality, lighting, vehicle models, damage severity, geography and annotation practices. The model may learn dataset-specific cues rather than damage features. Before any pilot, Bul Ins should verify whether suitable real claim photos and permissions exist and whether the proposed damage labels match its operational categories.

Part recognition and severity are separate tasks from damage-type classification. A single image may contain several damage instances, so cropping, object detection or multi-label approaches should be evaluated before committing to a production design.

1.4 Legal and regulatory environment

EU AI Act and insurance supervision

Under Annex III of the EU AI Act, high-risk insurance use cases explicitly include risk assessment and pricing for natural persons in life and health insurance. A motor-damage classification tool used solely for claims triage is not automatically classified as high-risk on that basis. Its exact regulatory classification must be reassessed if the intended use changes; being outside that specific high-risk category does not mean that the system is unregulated (European Union, 2024; European Commission, n.d.).

EIOPA recommends proportionate AI governance for insurance, including data governance, record-keeping, fairness, cyber security, explainability and human oversight. Bul Ins would need internal responsibility, review procedures and appropriate governance before production use (EIOPA, 2025).

GDPR, claims data and automated decisions

Photographs can contain personal data such as identifiable faces, registration plates or contextual details linked to a policyholder. If personal data are processed, GDPR principles such as purpose limitation, data minimisation, security and storage limitation apply. A lawful basis and controller/processor responsibilities must be determined before handling real claim data (European Union, 2016).

GDPR Article 22 concerns certain decisions based solely on automated processing with legal or similarly significant effects. Keeping the system as genuine decision support, with meaningful human review, helps avoid inappropriate reliance on automated outputs; legal assessment is still required for the specific workflow. A data protection impact assessment may be needed if processing is likely to pose high risks to individuals (European Union, 2016).

1.5 Ethical and social environment

Fairness, explainability and accountability

If a model performs poorly for certain damage classes, vehicle types or photo conditions, customers could experience delays or inconsistent

handling. Fairness must be evaluated using error rates across relevant categories, not assumed from aggregate accuracy.

The tool should show that its output is a prediction, provide uncertainty information, permit corrections and preserve a record of human review. A claims handler should be able to override the model without pressure to accept it. These safeguards reflect EIOPA guidance on fair and explainable insurance AI (EIOPA, 2025).

Misuse and cyber security

Photos may be altered, misleading, duplicated or uploaded without permission. The system could also expose sensitive claim records if access is poorly controlled. Recommended safeguards include access controls, audit logs, image integrity checks, secure retention policies and manual review for uncertain or suspicious cases. These are proposed mitigations, not existing Bul Ins controls.

1.6 Environmental and sustainability considerations

Model training, repeated experiments, image storage and inference consume electricity and computing resources. The environmental impact of this specific project has not been measured, so it would be misleading to claim a quantified carbon footprint or energy saving. Practical design choices include reusing preprocessed images, avoiding unnecessary retraining, testing smaller models before larger ones, deleting redundant image copies and selecting storage retention periods consistent with legal requirements. Any future sustainability assessment should record hardware, training duration, storage volume and electricity use. The wider energy implications of AI are discussed by the International Energy Agency (IEA, 2025).

1.7 Business value and future scenarios

Potential benefits include faster initial sorting, more consistent labels and reduced repetitive review. Potential costs include annotation work, infrastructure, staff training, model maintenance and errors that require manual correction. These benefits and costs have not yet been measured at Bul Ins. A pilot should compare processing time, error rate, staff workload and customer impact with the current workflow.

Repair-cost estimation is a possible later extension, but it needs structured data on vehicle brand, model, year, damaged part, severity, parts prices, labour, market value and historical paid claims. It would create higher stakes for customers and require a fresh legal, ethical and data assessment before development.

1.8 Findings, limitations and next steps

- The domain supports a narrow first objective: classify visible damage as an aid to human triage, not make compensation decisions.
- The first Bul Ins interview validates broader manual-work problems but not yet the specific photo-classification bottleneck.
- The public dataset and initial k-NN results are insufficient to claim production accuracy or fairness.
- Before a pilot: validate real-photo availability and permissions; confirm label definitions with inspectors; evaluate per-class performance; map human review and correction steps; complete privacy and security checks.
- Assess business impact and energy use empirically instead of claiming unmeasured benefits.

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